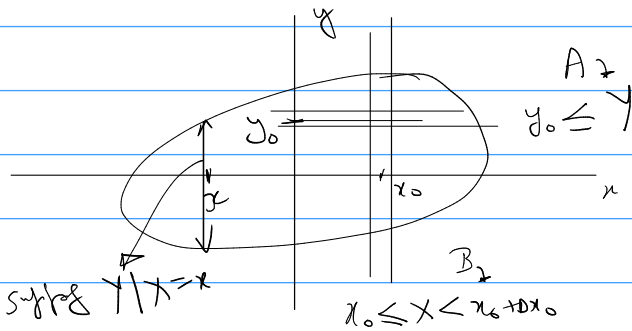
Sept 11 $X, Y \sim$ joint PDF f_{XY}

Marginals: f_X, f_Y

Conditional PDF:



$P(Y \approx y_0) = f_Y(y_0)\,\Delta y_0$

$$P(Y \approx y_0 \mid X \approx x_0) = P(A \mid B) = \frac{P(A \cap B)}{P(B)} \approx \frac{f_{XY}(x_0, y_0)\,\Delta x_0\,\Delta y_0}{f_X(x_0)\,\Delta x_0} = \frac{f_{XY}(x_0, y_0)}{f_X(x_0)}\,\Delta y_0$$

Conditional PDF

$Y|X=x_0$

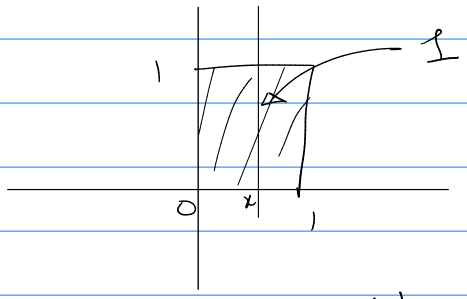
$$f_{Y|X=x_0}(y_0) = \frac{f_{XY}(x_0, y_0)}{f_X(x_0)}, \quad f_X(x_0) > 0$$

$f_{Y|X=x_0}(y_0)$ ↔ $f_{Y|X}(y_0|x_0)$

Conditional PDF of Y given X: $f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}$, $y: f_{XY}(x_0, y) > 0$ (y: variable, x: "constant")

" " " X " Y: $f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$, $x: f_{XY}(x, y) > 0$

Ex: (1)



$$f_{XY}(x,y) = \begin{cases} 1, & 0 < x, y < 1 \\ 0, & \text{else} \end{cases}$$

$$f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{else} \end{cases}$$

$$f_Y(y) = \begin{cases} 1, & 0 < y < 1 \\ 0, & \text{else} \end{cases}$$

$$f_{Y|X}(y|x) = \begin{cases} 1, & 0 < y < 1 \\ 0, & \text{else} \end{cases}$$

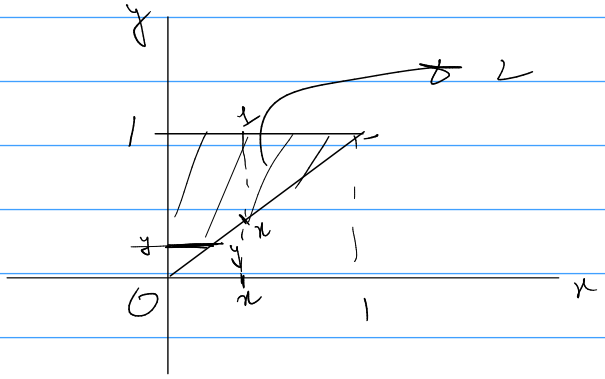
$$f_{X|Y}(x|y) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{else} \end{cases}$$

$Y|X=x \sim \text{Uniform}[0,1]$ $\quad Y \sim \text{Uniform}[0,1]$ $\quad X, Y$:

$X|Y=y \sim \text{Uniform}[0,1]$ $\quad X \sim \text{Uniform}[0,1]$ indep.

"conditional = marginal" for indep. RVs.

(2)



$$f_{X,Y}(x,y) = \begin{cases} 2, & 0<x<y<1 \\ 0, & \text{else} \end{cases}$$

$$f_X(x) = \begin{cases} 2(1-x), & 0<x<1 \\ 0, & \text{else} \end{cases}$$

$$f_{Y|X}(y|x) = \frac{1}{1-x} \quad x<y<1 \qquad f_Y(y) = \begin{cases} 2y, & 0<y<1 \\ 0, & \text{else.} \end{cases}$$

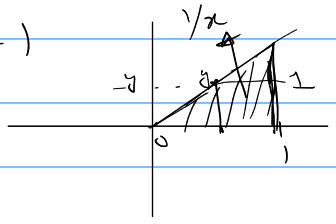
$$Y|X=x \sim \text{Uniform}[x,1]$$

$$X|Y=y \sim \text{Uniform}[0,y]$$

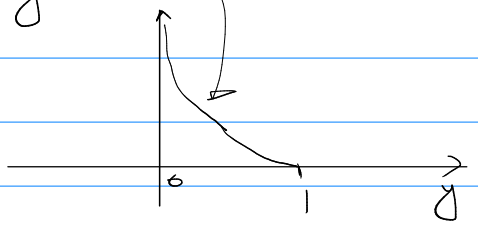
(3) $X \sim \text{Uniform}[0,1]$ $\qquad Y|X=x \sim \text{Uniform}[0,x]$

$$f_X(x) = \begin{cases} 1, & 0<x<1 \\ 0, & \text{else} \end{cases} \qquad f_{Y|X=x}(y) = \begin{cases} 1/x, & 0<y<x \\ 0, & \text{else} \end{cases}$$

$$f_{X,Y|x,y} = \begin{cases} \frac{1}{x}, & 0<y<x<1 \\ 0, & \text{else} \end{cases}$$

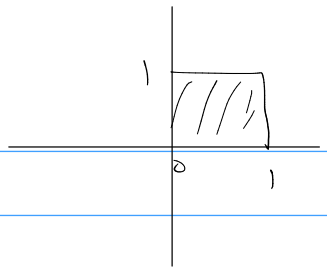


$$f_Y(y) = \int_{x=y}^{1} \frac{1}{x}\,dx = \log\frac{1}{y} = -\log y, \quad 0<y<1$$



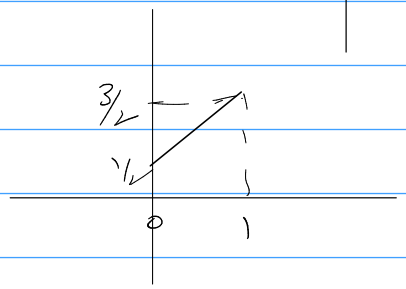
$$f_{X|Y=y}(x) = \begin{cases} \frac{-1}{x \log y} & y<x<1 \\ 0, & \text{else} \end{cases}$$

(3) $f_{XY}(x,y) = \begin{cases} x+y & , 0 < x,y < 1 \\ 0 & , \text{else} \end{cases}$



$f_X(x) = x + 1/2 \quad , 0 < x < 1$

$f_Y(y) = y + 1/2 \quad , 0 < y < 1$



$$f_{Y|X=x}(y) = \begin{cases} \frac{x+y}{x+1/2} & , 0 < y < 1 \\ 0 & , \text{else} \end{cases}$$

"x: "const"

(4) $f_{XY}(x,y) = \frac{1}{2\pi} e^{-x^2/2} e^{-y^2/2}$, all x,y

$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, $f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$

$f_{Y|X=x}(y) =$

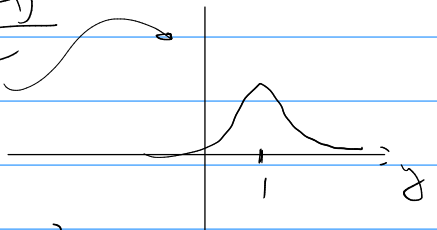
indep: easy....

Discrete & Continuous: X : discrete PMF $p_X(x)$, $x \in X$

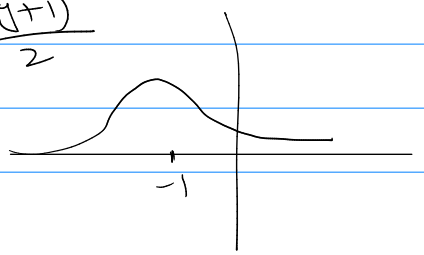
Y : Continuous PDF $f_Y(y)$ . . .

Ex: $X \sim \text{Uniform}\{-1, 1\}$

$Y \mid X = 1 \sim$ PDF $\frac{1}{\sqrt{2\pi}} e^{\frac{-(y-1)^2}{2}}$



$Y \mid X = -1 \sim$ PDF $\frac{1}{\sqrt{2\pi}} e^{\frac{-(y+1)^2}{2}}$



$$f_{Y}(y) = P(X=1) f_{Y|X=1}(y) + P(X=-1) f_{Y|X=-1}(y)$$

$$P(X=1 \mid Y=y) = \frac{f_{Y|X=1}(y) \cdot P(X=1)}{f_{Y}(y)}$$